

Module 3 Activity 4: Using the Update Cursor

Purpose:

Use the **UpdateCursor()** function with the **AddField_management** tool to update values in a new field.

Instructions:

You can use any Python editor for this exercise.

Part 1:

1. Import the **arcpy** module
2. Set the workspace
3. Identify a feature class you want to work with
4. Create a new field using the **AddField_management** function
5. Name the field **REVIEW** and set the field length to 20
6. Inform the user to tell them that the new field has been added by adding a **print** statement

The syntax for the **AddField_management** function is below. Consult the tool help for details on each parameter.

AddField_management (in_table, field_name, field_type, {field_precision}, {field_scale}, {field_length}, {field_alias}, {field_is_nullable}, {field_is_required}, {field_domain})

```
Python
#import the arcpy module
import arcpy
#set the workspace environment
from arcpy import env
env.workspace = r"C:\GEOS456\Data.gdb"
#Choose a feature class to work with
fc = "Crimes2009"
#add a new field to the table and specify the field length
arcpy.AddField_management("Crimes2009","REVIEW","TEXT","","","20")
#tell the user that the field has been added
print ("REVIEW field has been added to the Crimes2009 table")
#the use of GetMessages could be useful as well
```

At this point, you can run the code. Once it's run, a new field called **REVIEW** will be added to the feature class attribute table and you will be notified in the Python Interpreter that the creation of the field has been successful. For the purposes of this activity, we will not run the code until it is complete.

7. Create the **UpdateCursor** inside a variable called **UCursor**
8. Reference the feature class in the **UpdateCursor** syntax and specify the fields you want to work with. Fields are a required parameter in the **UpdateCursor** and are referenced using square brackets **[]**

For the purposes of this activity, we want different values in the newly created **REVIEW** field based on the values of the **OFFDESC** field, which is already in existence. See screenshot on the next page

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#add a new field to the table and specify the field length
arcpy.AddField_management("Crimes2009","REVIEW","TEXT","","","20")
#tell the user that the field has been added
print ("REVIEW field has been added to the Crimes2009 table")
#the use of GetMessages could be useful as well
#use the update cursor and identify the fields to work with
UCursor = arcpy.da.UpdateCursor(fc,["OFFDESC","REVIEW"])
```

NOTE: Remember that any field inside a cursor can be indexed. In this case, the **OFFDESC** is an index value of **0** and the newly created **REVIEW** field is an index value of **1**.

9. Begin a **for** loop that will contain an **if/else** statement
10. Use the **updateRow** method to update the values of the new field

The **if/else** statement will allow us to use a conditional statement to specify what new values will be created in the **REVIEW** field based on specific attribute information from the **OFFDESC** field

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arcpy.AddField_management("Crimes2009","REVIEW","TEXT","","","20")
#tell the user that the field has been added
print ("REVIEW field has been added to the Crimes2009 table")
#the use of GetMessages could be useful as well
#use the update cursor and identify the fields to work with
UCursor = arcpy.da.UpdateCursor(fc,["OFFDESC","REVIEW"])
#start a for loop with a variable referencing the UCursor
for row in UCursor:
    #use an IF/ELSE statement to specify what values
    #the new field will contain
    if row[0] == "ACCIDENTAL DEATH":
        row[1] = "Pending"
    elif row[0] == "ACCIDENTAL SHOOTING":
        row[1] = "Pending"
    elif row[0] == " ":
        row[1] = "Insufficient Info"
    else:
        row[1] = "Closed"
    #use the updateRow method to update the values in the new field
    UCursor.updateRow(row)
```

NOTE: The *if/else* statement is referencing the index position of the fields in the *UpdateCursor* function

11. Tell the user that the values have been added using the ***print*** statement
12. Use the ***del*** function to remove any locks on the data

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        row[1] = "Pending"
    elif row[0] == " ":
        row[1] = "Insufficient Info"
    else:
        row[1] = "Closed"
    #use the updateRow method to update the values in the new field
    UCursor.updateRow(row)
#tell the user that values have been added
print ("REVIEW field has been updated")
#remove any locks created by the cursor objects
del UCursor
del row
```

13. Run the code

14. Examine your results

Crimes2009

Field:	Add	Calculate	Selection:	Select By Attributes	Zoom To	Switch	Clear	Delete	Copy			
SPLITDT	SPLITM	HR	DOW	SHIFT	OFFCODE	OFFDESC	ARCODE	ARCODE2	ARCTYPE	XNAD83	YNAD83	REVIEW
60883	1/17/2009	12/30/1899	21	Wed	B					2129728.38	13773721.97	Insufficient Info
60884	1/17/2009	12/30/1899	20	Wed	B	000030	MIS CHILD	OTHER	OTHER	2071405.38	13736986.97	Closed
60885	1/17/2009	12/30/1899	23	Wed	C	131352	SIMPLE ASSLT	YFAMVIOL	FAMVIOL	2145463.38	13703096.97	Closed
60886	1/17/2009	12/30/1899	23	Wed	C	290222	DAMAGE PROP-PRIVAT	YVANDAL	VANDAL	2084959.38	13714626.97	Closed
60887	1/17/2009	12/30/1899	23	Wed	C	359923	DANGEROUS DRUGS	YDRUGARR	DRUGARR	2140318.38	13751376.97	Closed
60888	1/17/2009	12/30/1899	23	Wed	C	131302	SIMPLE ASSLT	YASSAULT	ASSAULT	2072620.38	13699899.97	Closed
60889	1/17/2009	12/30/1899	23	Wed	C	356205	MARIJUANA-POSSESS	YDRUGARR	DRUGARR	2072878.38	13699913.97	Closed
60890	1/17/2009	12/30/1899	23	Wed	C	480107	RESISTING OFFICER	YOBSTRUCT	OBSTRUCT	2094313.38	13722396.97	Closed
60891	1/17/2009	12/30/1899	23	Wed	C	120103	ROBBERY-BUSINESS-GL	YAROBUS	AROBUS	2083558.38	13736351.97	Closed
60892	1/18/2009	12/30/1899	23	Thu	C					2104624.38	13732391.97	Insufficient Info
60893	1/18/2009	12/30/1899	3	Thu	C	290221	DAMAGE PROP-PRIVAT	YVANDAL	VANDAL	2117162.38	13698864.97	Closed

0 of 182,310 selected

Filters: 100%